Over time, drones are expected to support resource prospecting, construction of extraterrestrial infrastructure, and logistical transport in support of sustained human presence in space.

As exploration shifts toward multi-planetary habitation and industrial utilization, drones will serve as autonomous partners, capable of performing hazardous, repetitive, or high-precision tasks. Their ability to operate in extreme environments marks a new phase in off-Earth robotics one where the sky, even on other worlds, becomes an active domain of scientific discovery and human expansion.

9.7.1 The potential role of drones in extraterrestrial exploration and planetary research

Drones offer significant advantages for extraterrestrial exploration because they can move freely across planetary surfaces and through atmospheric environments, where conventional rovers and land vehicles face severe limitations. Planetary terrains such as crater rims, canyon walls, volcanic fields, ice cliffs, and cavern openings are often inaccessible or hazardous for wheeled systems, greatly restricting the range and speed of surface exploration. In contrast, aerial and hopping drones can quickly traverse difficult landscapes, collect data from multiple locations in a short period, and provide high-resolution mapping across areas that would take rovers weeks or months to cover.

Equipped with multispectral cameras, atmospheric sensors, ground-penetrating imaging systems, and mineral detection instruments, drones can play a critical role in studying planetary geology, climate evolution, surface composition, and Astro biological potential. They can scout landing sites for future missions, locate promising regions for sample collection, and identify natural shelters such as lava tubes that could protect future astronauts from radiation. Because communication delays in deep space limit real-time